

Mathematical Background for NTRU TowerSolver

PRITAM CHANDRA

This paper attempts to provide the theoretical background necessary for understanding the TowerSolver_{*n,q*} algorithm to obtain solutions of the NTRU equations, as proposed in [this](#) paper by Thomas Pornin and Thomas Prest.

1 Preliminaries and notations

1. ϕ_n The polynomial $\phi_n(x) = x^n + 1$.
2. ς_n The complex number $\varsigma_n = e^{\frac{i\pi}{n}}$ is a root of ϕ_n . Note the set of the roots of ϕ_n is $\{\varsigma_n^j : j = 0, 1, \dots, n-1\}$.
3. $\mathbb{Q}_\varphi, \mathbb{Z}_\varphi$ For a given polynomial $\varphi(x) \in \mathbb{Q}[x]$ (or $\mathbb{Z}[x]$), $\mathbb{Q}_\varphi$ (or $\mathbb{Z}_\varphi$) denotes the quotient field $\frac{\mathbb{Q}[x]}{(\varphi(x))}$ (or the quotient ring $\frac{\mathbb{Z}[x]}{(\varphi(x))}$).
4. $[n]$ The set $[n] = \{0, 1, \dots, n-1\}$.
5. $\{\varphi(j)\}_{[n]}$ For any function φ , the set $\{\varphi(j)\}_{[n]} = \{\varphi(j) : j \in [n]\}$.
6. $(\varphi(j))_{[n]}$ The ordered tuple $(\varphi(j))_{[n]} = (\varphi(j) : j \in [n])$
7. $*$ product The $a(x) * b(x)$ is the convolution of two polynomials without any modular reduction. Products specific to a given field will be denoted without a symbol. For example, for a, b in $\mathbb{Q}_\varphi$, the product ab is given as $ab(x) = a(x) * b(x) \bmod \varphi$.

Sometimes ς_n or ϕ_n can be used without the subscript n as ς or ϕ , when n is clear from the context.

For an $n-1$ degree polynomial denoted by a notation a , it is automatically assumed that $a_0 + a_1x + \dots + a_{n-1}x^{n-1}$. Moreover, this notation a can be interchangeably used to represent the n -length vector a whose components are automatically assumed to be $(a_0, \dots, a_{n-1})$.

Any notation A used to denote a linear map A will be interchangeably used to represent its matrix representation (with respect to the standard basis unless mentioned otherwise), and also the set of its column vectors.

2 Resultants

For $f, g \in \mathbb{Q}[x]$ we define the resultant $\text{res}(f, g)$ of f and g as the determinant of the $\mathbb{Q}$ -linear map

$$\Phi_{f,g}: \mathcal{P}_k \times \mathcal{P}_m \rightarrow \mathcal{P}_{m+k} \text{ where } (s, t) \mapsto sf + tg.$$

Equivalently, resultants can be defined in terms of the roots $\{\alpha_j\}_{[m]}$ of f and the roots $\{\beta_j\}_{[k]}$ of g as follows:

$$\text{res}(f, g) = f_0^m \prod_{j=0}^m g(\alpha_j) = (-1)^{mk} g_0^k \prod_{j=0}^k f(\beta_j). \quad (1)$$

3 Field norm

Let K/L be a field extension for the fields K and L . For $a \in K$, the L -linear multiplication operator on K corresponding to a , denoted by $T_{K/L}(a)$, is the unique linear map with the following properties:

$$T_{K/L}(a) \cdot b = ab \text{ for all } b \in K, \text{ and } T_{K/L}(l) = l \text{ for all } l \in L.$$

Then, we denote the field norm of a from K to L by $N_{K/L}(a)$, and define it as

$$N_{K/L}(a) = \det T_{K/L}(a).$$

Note that the map $T_{K/L}$ being linear is represented with an n -dimensional square matrix, where $n = [K:L]$, the index of L in K .

Further, it is known that field norm is transitive over a tower of extensions $K/L/M$. That is

$$N_{K/M}(a) = N_{L/M}[N_{K/L}(a)]. \quad (2)$$

A concrete proof of the above fact can be found [here](#).

4 Field norm from $\mathbb{Q}_{\phi_{2^n}}$ to $\mathbb{Q}_{\phi_n}$

Let $K = \mathbb{Q}_{\phi_{2^n}} = \frac{\mathbb{Q}[x]}{(\phi_{2^n})}$ and $L = \mathbb{Q}_{\phi_n} = \frac{\mathbb{Q}[y]}{(\phi_n)}$. For notational convenience, we distinguish the indeterminates of K and L by x and y . We also denote $T_{K/L}$ and $N_{K/L}$ by T and N respectively.

Now, K is a field extension of L by the injective homomorphism $y \mapsto x^2$. Let a be an arbitrary element of K . Then

$$\begin{aligned}
 a(x) &= a_0 + a_1x + \cdots + a_{2n-1}x^{2n-1} \\
 &= (a_0 + a_2x^2 + \cdots + a_{2n}x^{2(n-1)}) \\
 &\quad + x(a_1 + a_3x + \cdots + a_{2n-1}x^{2(n-1)}) \\
 &= a_e(x^2) + x a_o(x^2) \\
 &= a_e(y) + x a_o(y)
 \end{aligned}$$

These leads us to the following definition.

Definition 1. Every $a \in K$ can be represented with two elements a_e, a_o of L . The element $a_e = (a_{2j})_{[n]}$ is called the even-part of a , and $a_o = (a_{2j+1})_{[n]}$ is called the odd-part of a . In particular, we have

$$a = (a_e, a_o), \text{ such that } a(x) = a_e(y) + x a_o(y), \text{ where } y = x^2.$$

Simiarly, using the same representation for $b \in K$ we can obtain the product

$$\begin{aligned}
 ab(x) &= a(x)b(x) \\
 &= [a_e(y) + x a_o(y)] [b_e(y) + x b_o(y)] \\
 &= a_e(y)b_e(y) + x^2 a_o(y)b_o(y) + x[a_o(y)b_e(y) + a_e(y)b_o(y)] \\
 &= (a_e b_e + y a_o b_o)(y) + x(a_o b_e + a_e b_o)(y)
 \end{aligned}$$

Representing the above result as denoted in definition 1 we have

$$ab = (a_e b_e + y a_o b_o, a_o b_e + a_e b_o).$$

With this we can immediately determine the L -linear multiplication operator $T(a)$ on K corresponding to the element $a \in K$, where

$$T(a): (b_e, b_o) \mapsto (a_e b_e + y a_o b_o, a_o b_e + a_e b_o).$$

From here we can obtain the matrix representation of $T(a)$ by obtaning its effect on the standard basis vectors $(1, 0)$ and $(0, 1)$ as

$$T(a) = \begin{bmatrix} a_e & y a_o \\ a_o & a_e \end{bmatrix}.$$

Finally, $N(a) = \det T(a) = a_e^2 - y a_o^2$. Further observe that

$$\begin{aligned}
 N(a)(y) &= a_e^2(y) - y a_o^2(y) \\
 &= a_e^2(x^2) - x^2 a_o^2(x^2) \\
 &= [a_e(x^2) + x a_o(x^2)] [a_e(x^2) - x a_o(x^2)] \\
 &= a(x) a(-x).
 \end{aligned}$$

We club all the results above into the following theorem.

Theorem 2. *With K and L as in this section, for $a \in K$, the matrix representation of the L -linear multiplication operator on K corresponding to a is given by*

$$T(a) = \begin{bmatrix} a_e & y a_o \\ a_o & a_e \end{bmatrix},$$

where $y = x^2$. Subsequently, the field norm of a from K to L is given by

$$N(a) = a_e^2 - y a_o^2 \text{ and } N(a)(y) = a(x) a(-x).$$

5 Skew circulant matrices and their eigenvalues

Consider an n -length real vector a . Then the skew circulant matrix defined by a is denoted by $C'(a)$ and is expressed as

$$C'(a) = \begin{bmatrix} a_0 & a_1 & a_2 & \dots & a_{n-1} \\ -a_{n-1} & a_0 & a_1 & \dots & a_{n-2} \\ -a_{n-2} & -a_{n-1} & a_0 & \dots & a_{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -a_1 & -a_2 & -a_3 & \dots & a_0 \end{bmatrix}. \quad (3)$$

Let $e = (0, 1, 0, \dots, 0)$. Then define τ as

$$\tau = C'(e) = \begin{bmatrix} 0 & I_{n-1} \\ -1 & 0 \end{bmatrix}.$$

We call τ as the *fundamental skew-circulant matrix of dimension n* . We will see a good reason behind this name later. For now we make the following proposition.

Proposition 3. *The matrix τ has the following properties.*

$$i. \quad \tau^k = \begin{bmatrix} 0 & I_{n-k} \\ -I_k & 0 \end{bmatrix}$$

ii. *The characteristic polynomial of τ is $(-1)^n(x^n + 1)$.*

Proof. For (i.) we proceed by induction. The base case, for $k = 1$, follows directly from the definition of τ .

Assume the proposition holds for the case k . Then

$$\begin{aligned}\tau^{k+1} &= \tau^k \cdot \tau \\ &= \begin{bmatrix} 0 & I_{n-k} \\ -I_k & 0 \end{bmatrix} \begin{bmatrix} 0 & I_{n-1} \\ -1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & I_{n-k-1} \\ -I_{k+1} & 0 \end{bmatrix}\end{aligned}$$

For the last step notice that for any matrix A , the first column of $A\tau$ is the negative of the last column of A , and the last $n-1$ columns of $A\tau$ are the first $n-1$ columns of A . This completes the induction.

Using (i.) it immediately follows that

$$\tau^n = \begin{bmatrix} 0 & I_0 \\ -I_n & 0 \end{bmatrix} = I_n \implies \tau^n + I = 0.$$

Now, the characteristic polynomial of an n -dimensional matrix is the unique monic polynomial of degree n . We obtain (ii.) by coupling this with the Cayley-Hamilton theorem which states that every matrix satisfies its characteristic polynomial. In particular, we get that the characteristic polynomial of τ^n is either $x^n + 1$ or its negative. The additional factor of $(-1)^n$ can be understood by noting that each of the diagonal elements of the matrix $(\tau - xI_n)$ is $-x$, therefore the leading term of the characteristic polynomial of τ must be $(-x)^n$. $\square$

Corollary 4. *The eigenvalues of τ are the roots of its characteristic polynomial $x^n + 1$. Precisely,*

$$\sigma(\tau) = \{\zeta^{2j+1}\}_{[n]}.$$

Using (3.i.) and the definition of skew circulant matrices, we can immediately obtain the next corollary.

Corollary 5. *The skew circulant matrix defined by the vector $a = (a_0, \dots, a_{n-1})$ is a polynomial in τ . Precisely,*

$$\mathcal{C}'(a) = a_0 + a_1\tau + \dots + a_{n-1}\tau^{n-1}.$$

Corollary 5 is the primary reason why call τ as the fundamental skew circulant matrix of dimension n . Now, because $\sigma(\tau)$ has n distinct elements, the matrix τ is diagonalizable in its eigenbasis. Or, there exists an invertible matrix V whose columns are the eigenvectors of τ , and a diagonal matrix D whose diagonal entries are the eigenvalues of τ , such that $\tau = VDV^{-1}$. Then

$$\begin{aligned}V^{-1}\mathcal{C}'(a)V &= a_0 + a_1(V^{-1}\tau V) + \dots + a_{n-1}(V^{-1}\tau V)^{n-1} \\ &= a_0 + a_1D + \dots + a_{n-1}D^{n-1} \\ &= \text{diag}\{a(\zeta^{2j+1})\}_{[n]},\end{aligned}$$

where $a(x)$ is taken as the polynomial corresponding to the vector a . From this we obtain yet another corollary.

Corollary 6. *All skew circulant matrices share the same eigenvectors, and are hence diagonalized in the same basis. Moreover, for a skew circulant matrix $\mathcal{C}'(a)$ defined by the vector a , its spectrum is given by*

$$\sigma[\mathcal{C}'(a)] = \{a(\zeta^{2j+1})\}_{[n]}.$$

In the next section we see the relationship of skew circulant matrices to our convolution field $\mathbb{Q}_\phi$.

6 Skew circulant matrices in context of the field $\mathbb{Q}_\phi$

Let $K = \mathbb{Q}_\phi$. For $a \in K$, we have seen that the $\mathbb{Q}$ -linear multiplication operator on K corresponding to a is denoted by $T_{K/\mathbb{Q}}(a)$, where by definition

$$T_{K/\mathbb{Q}}(a) \cdot b = ab \text{ for all } b \in K. \quad (4)$$

For the matrix representation of this map consider its action on e_j , the j -th standard basis vector. That is

$$a e_j(x) = a(x) * e_j(x) \bmod \phi = a(x) * x^j \bmod \phi \text{ for } j \in [n].$$

Thus, the j -th column of $T_{K/\mathbb{Q}}(a)$ is the vector $(-a_{n-j}, \dots, -a_{n-1}, a_0, \dots, a_{n-j+1})$. This vector precisely being the j -th row of $\mathcal{C}'(a)$ sets the ground for the following lemma.

Lemma 7. *For any $a \in K$, the $\mathbb{Q}$ -linear multiplication operator on K corresponding to a is given by the skew circulant matrix $\mathcal{C}'(a)^T$. That is,*

$$T_{K/\mathbb{Q}}(a) = \mathcal{C}'(a)^T.$$

After this we are all set to establish the theorem crucial to this discussion.

Theorem 8. *Let $K = \mathbb{Q}_\phi$, and let $a \in K$. Then $N_{K/\mathbb{Q}}(a) = \text{res}(\phi, a)$.*

Proof. For any $a \in K$ the following deductions take us through the proof.

$N_{K/\mathbb{Q}}(a)$	$= \det T_{K/\mathbb{Q}}(a)$	by def of $N_{K/\mathbb{Q}}$
	$= \det \mathcal{C}'(a)^T$	lemma 7
	$= \det \mathcal{C}'(a)$	properties of determinants
	$= \prod_{j=0}^{n-1} a(\zeta^{2j+1})$	det is product of eigenvalues
	$= \text{res}(\phi, a)$	by def of resultant (1) $\square$

From here we constitute the next corollary, which is the primary result involved in solving the NTRU equations.

Corollary 9. *Let $K = \mathbb{Q}_{\phi_{2n}}$ and $L = \mathbb{Q}_{\phi_n}$. Then for any $a \in K$,*

$$\text{res}(\phi_{2n}, a) = \text{res}[\phi_n, N_{K/L}(a)].$$

Proof. The result is obtained by directly applying theorem 6 on the equation of field norm transitivity $N_{K/\mathbb{Q}}(a) = N_{L/\mathbb{Q}}[N_{K/L}(a)]$. $\square$ ✠ see page 7

7 The NTRU problem and its solution in $\mathbb{Z}$

Problem. *Let q and l be integers (non-negative) and let $n = 2^l$. Then given $f, g \in \mathbb{Z}_{\phi_n}$, we are required to find $F, G \in \mathbb{Z}_{\phi_n}$ such that the NTRU equation holds. That is*

$$fG - gF = q. \quad (5)$$

When $n = 1$, $\mathbb{Z}_{\phi_n} = \mathbb{Z}$. Let $f, g \in \mathbb{Z}$, and say $\delta = \gcd(f, g)$. If $\delta \nmid q$ we conclude that the NTRU equations for this choice of f and g cannot be solved. Otherwise, by extended Euclidean algorithm we find $s, t \in \mathbb{Z}$ such that

$$fs + gt = \delta.$$

We now pick $F, G \in \mathbb{Z}$ as follows and verify that they are indeed a solution.

$$F = tq/\delta \text{ and } G = sq/\delta. \quad (6)$$

$$fG - Fg = f \frac{sq}{\delta} + g \frac{tq}{\delta} = \frac{q}{\delta}(fs + gt) = q.$$

8 Converting NTRU solutions in $\mathbb{Z}_{\phi_n}$ to solutions in $\mathbb{Z}_{\phi_{2n}}$

Let $K = \mathbb{Q}_{\phi_{2n}}$ and $L = \mathbb{Q}_{\phi_n}$. For easing notations denote $N_{K/L}$ by N .

Let $f, g \in \mathbb{Z}_{\phi_{2n}}$. Let $f' = N(f)$ and $g' = N(g)$. Then f' and g' are determinants of matrices whose entries are integral polynomials. Hence $f', g' \in \mathbb{Z}_{\phi_n}$.

Assume we can solve the NTRU equations for f' and g' . That is, we know $F', G' \in \mathbb{Z}_{\phi_n}$ such that

$$f'G' - g'F' = q.$$

Then pick $F, G \in \mathbb{Z}_{\phi_{2n}}$ as follows.

$$\begin{aligned} F(x) &= g(-x) F'(x^2) \\ G(x) &= f(-x) G'(x^2). \end{aligned} \tag{7}$$

We now verify that F, G solve the NTRU equations defined by f, g .

$$\begin{aligned} (fG - gF)(x) &= f(x) G(x) - g(x) F(x) \\ &= f(x) f(-x) G'(x^2) - g(x) g(-x) F'(x^2) \\ &= f'(x^2) G'(x^2) - g'(x^2) F'(x^2) \\ &= (f'G' - g'F')(x^2) \\ &= q. \end{aligned}$$

9 The TowerSolver $_{n,q}$ algorithm

The pieces of the TowerSolver $_{n,q}$ algorithm already lies in the discussions above. We combine them here to provide the final algorithm.

Algorithm TowerSolver $_{n,q}(f, g)$

input: $f, g \in \mathbb{Z}_{\phi_n}$ where n is a power of 2; a non-negative integer q

output: $F, G \in \mathbb{Z}_{\phi_n}$ satisfying the NTRU equations with f and g .

if $n = 1$,

1. Use extended euclidean algorithm to find $\delta = \gcd(f, g)$ and s and t such that $fs + gt = \delta$.
2. If $\delta \nmid q$, abort the program.
3. Follow (6) and set $F = {}^tq/\delta$ and $G = {}^sq/\delta$.
4. Return F, G .

Otherwise,

1. Follow Theorem 2 and set $f' = f_e^2 - x f_o^2$ and $g' = g_e^2 - x g_o^2$.
2. Call TowerSolver $_{n/2,q}(f', g')$ to obtain F' and G' .
3. Follow (7) to set $F(x) = g(-x) F'(x^2)$ and $G(x) = f(-x) G'(x^2)$.
4. Return F, G .

✱ Following our discussion in 4 we can deduce corollary 9 without invoking the transitivity of field norm. This can be done as follows.

$\text{res}(\phi_{2n}, a) = \prod_{j=0}^{2n-1} a(\varsigma_{2n}^{2j+1})$	by def of res (1)
$= \prod_{j=0}^{n-1} a(\varsigma_{2n}^{2j+1}) a(\varsigma_{2n}^{2(n+j)+1})$	Rearrange
$= \prod_{j=0}^{n-1} a(\varsigma_{2n}^{2j+1}) a(-\varsigma_{2n}^{2j+1})$	$\varsigma_{2n}^2 = -1$
$= \prod_{j=0}^{n-1} N_{K/L}(a)[(\varsigma_{2n}^{2j+1})^2]$	Theorem 2
$= \prod_{j=0}^{n-1} N_{K/L}(a)[(\varsigma_{2n}^2)^{2j+1}]$	Exchange the powers
$= \prod_{j=0}^{n-1} N_{K/L}(a)(\varsigma_n^{2j+1})$	$\varsigma_{2n}^2 = \varsigma_n$
$= \text{res}[\phi_n, N_{K/L}(a)]$	by def of res (1)